Project Report Format

1. **INTRODUCTION**

1.1

- Project Overview : The project "EV Market Analysis Using Tableau" aims to analyze the global Electric Vehicle market data (1.5M+ records) to identify trends in sales by category, region, and manufacturing country. We used Snowflake for data warehousing and Tableau for visualization. The dashboard shows SUV dominates with 174M sales, South Asia leads region-wise with 210M sales, and India leads manufacturing.

1.2

Purpose: To provide data-driven insights for EV manufacturers and investors to understand market demand, regional growth, and consumer preferences to make better business decisions.

2. **IDEATION PHASE**

2.1

Problem Statement: Lack of clear visualization of EV market trends leads to poor decision-making. Companies don't know which category sells more, which region is booming, and where to invest.

2.2

Empathy Map Canvas: Users: EV Companies, Investors, Government. Thinks: Where is market growing? Feels: Confused due to raw data. Says: Need simple dashboard. Does: Checks Excel sheets.

2.3

Brainstorming: Ideas - Use Snowflake + Tableau, Create 8 charts, 1 dashboard, 1 story with filters for Category, Region, Country.

3. **REQUIREMENT ANALYSIS**

3.1

Customer Journey map: Data Collection -> Snowflake Loading -> Cleaning -> Tableau Connection -> Visualization -> Dashboard -> Insights.

3.2

Solution Requirement: Hardware: Laptop 8GB RAM. Software: Snowflake, Tableau Public/Desktop, Excel. Dataset: EV_Market.xlsx (1.5M rows).

3.3

Data Flow Diagram: Excel Dataset -> Snowflake (EV_SALES table) -> Tableau Data Source -> Worksheets -> Dashboard -> Story -> Tableau Public Link.

3.4

Technology Stack: Frontend: Tableau Dashboard & Story, Backend: Snowflake Data Warehouse, Language: SQL, Tool: Tableau Prep for cleaning, Platform: Tableau Public.

4. PROJECT DESIGN

4.1

Problem Solution Fit: Problem is unstructured EV data, Solution is interactive Tableau dashboard with filters and story points that answers all market questions in one click.

4.2

Proposed Solution: Develop a Tableau dashboard with 8 visualizations showing Category-wise, Region-wise, Country-wise sales, Battery capacity trends, with live filters and publish to Tableau Public.

4.3

Solution Architecture: Data Layer (Excel -> Snowflake) -> Processing Layer (SQL queries, Calculated fields like YoY Growth) -> Visualization Layer (Tableau Worksheets, Dashboard, Story) -> User Layer (Tableau Public URL).

5. PROJECT PLANNING & SCHEDULING

5.1

Project Planning: Week 1: Data Collection & Cleaning, Week 2: Snowflake setup & SQL queries, Week 3: Tableau visualizations, Week 4: Dashboard & Story design, Week 5: Performance Testing & Final Report.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1

Performance Testing: Snowflake: Query time improved from 1.42 sec to 0.68 sec after filters (52% improvement). Tableau: Dashboard load time 1.8 sec, 8 visualizations working, all filters tested successfully. Data Rendered: 1,550,704 rows.

7. RESULTS

7.1 Output Screenshots :
https://public.tableau.com/app/profile/ayush.anwalia/viz/EV_Market_Analysis_Ayush/Dashboard1

Story Link:

https://public.tableau.com/app/profile/ayush.anwalia/viz/EV

_Market_Analysis_Ayush/Dashboard1

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8. ADVANTAGES & DISADVANTAGES :

Easy to understand market trends, Interactive filters, Real-time insights, Supports business decisions, Cloud-based.

Disadvantages: Requires internet for Tableau Public, Large dataset may slow on low RAM, Limited to historical data only.

9. CONCLUSION :

The project successfully analyzed EV market and found SUV category and South Asia region are future leaders. Tableau dashboard provides actionable insights for EV stakeholders

10. FUTURE SCOPE :

The project successfully analyzed EV market and found SUV category and South Asia region are future leaders. Tableau dashboard provides actionable insights for EV stakeholders

11. APPENDIX Source Code(if any) :

Dataset Link:

Dataset Name: EV_Market.xlsx

Records: 1,550,704 Rows, 5 Columns (Category, Market_Region, Manufacturing_Country, Sales, Battery_Capacity)

GitHub & Project Demo Link

GitHub Repository: https://github.com/ayush-anwalia/-EV-Market-Analysis-Using-Tableau

Tableau Public Dashboard (Demo): https://public.tableau.com/app/profile/ayush.anwalia/viz/EV_Market_Analysis_Ayush/Dashboard1

Project Demo Link / Story Link:

https://public.tableau.com/app/profile/ayush.anwalia/viz/EV_Market_Analysis_Ayush/Dashboard1